ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 9



UNIT 2
Electronic Spreadsheets

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Say what an electronic spreadsheet, a workbook and a worksheet are.
- [ ] Use the common tools: New, Open, Save, Print, Undo, Cut, Copy, Paste ...
- [ ] Name the Name box, the Formula Bar, columns, rows, cells and cell ranges.
- [ ] Write cell addresses such as G7 and cell ranges such as D5:E10.
- [ ] Format cells with the formatting, alignment and number tools.
- [ ] Write formulas with = + - * / and use SUM, AVERAGE, MAX and MIN.
- [ ] Show data in column, line, pie, bar, area and scatter charts.

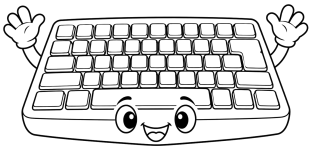
THREE RULES FOR THE COMPUTER ROOM

- **Always** save your work often, with your name in the file name.
- **Never** eat or drink near the computers.
- **Always** ask your teacher before you install or delete any software.

HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the practicals (activities 2.1 - 2.10) on a computer. Last you **write** the exercises in this book.

Colour the small pictures when you finish your work early.



Colour me!

THE WORDS YOU WILL LEARN

Spreadsheet	Workbook	Worksheet	Cell
Column	Row	Cell address	Cell range
Name box	Formula Bar	Print Preview	Undo / Redo
Font	Fill colour	Borders	Wrap Text
Merge and Center	Currency	Percentage	Formula
Function	SUM	AVERAGE	Chart

"Rows, columns and a little curiosity: the spreadsheet does the rest."

- THE KING OF ICT

CHAPTER 1

Spreadsheets and Their Common Tools

? MR. PERERA'S BIG BOOK

"Every evening I add the sales of my shop in this big ruled book. One small mistake, and I must add everything again!"

There is a better way: an **electronic spreadsheet**. It keeps the numbers in rows and columns and does the adding for you, quickly and correctly.



2.1 What are spreadsheets?

A spreadsheet is an **electronic version** of the paper based **accounting sheets** used by accountants in the past.

Day	Rs.
Monday	2,450
Tuesday	3,100
Wednesday	2,800
Thursday	3,350
Friday	2,700
Total	**14,400**

the same data

in a spreadsheet

B7 | *fx* | =SUM(B2:B6)

	A	B
1	**Day**	**Amount (Rs.)**
2	Monday	2450
3	Tuesday	3100
4	Wednesday	2800
5	Thursday	3350
6	Friday	2700
7	**Total**	**14400**

The money of the school canteen for one week. On paper the total is added by hand; in the spreadsheet the formula =SUM(B2:B6) adds it.

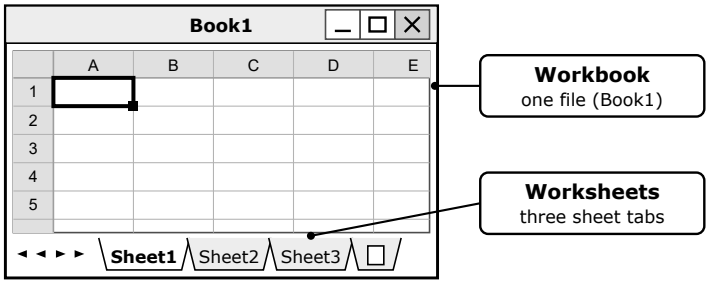
An **electronic spreadsheet** is an interactive computer application for the **organization, analysis and storage of data** in a **tabular form** (a table). Data are entered in the **cells** of the table.

i A SPREADSHEET GIVES US

- functions and formulas (calculations)
- sorting (putting data in order)
- charting (showing data in charts)

MS Excel and LibreOffice Calc are two spreadsheet programs that we can use in the computer lab.

2.2 Workbooks and worksheets



A spreadsheet file is a **workbook**. Workbooks can be used for document creation.

A workbook may contain **multiple worksheets**. Each worksheet has a **sheet tab**. Click a tab to see that worksheet.

2.3 The common tools

Let us learn about the tools of a spreadsheet package. These **common tools** are in most programs, in word processing too (textbook page 23):

Icon	Tool	What it does
	**New**	To create a new workbook.
	**Open**	To open a saved document.
	**Save**	To save a document.
	**Print Preview**	To see how a document will look if printed.
	**Print**	To print a document.

Icon	Tool	What it does
	**Undo**	To cancel a performed action.
	**Redo**	To do an undone action again.
	**Cut**	To remove an item from its current location.
	**Copy**	To copy letters, images, etc.
	**Paste**	To paste a copied or cut item.
	**Spelling**	To check the accuracy of spelling and grammar.

DO IT - PRACTICAL 1 (WORKBOOK ACTIVITY 2.1)

A workbook and its worksheets. **Ask your teacher first.**

1	Start MS Excel or LibreOffice Calc. A new **workbook** opens.
2	Find the **sheet tabs** at the bottom. Each tab is one **worksheet**.
3	Click each sheet tab. Add a worksheet with the **Insert Worksheet** tab.
4	Click **Sheet1**, type your name in the first cell and press **Enter**.

DO IT - PRACTICAL 2 (WORKBOOK ACTIVITY 2.2)

1	Click **New**. Type the names of five friends, one under the other.
2	Click **Save**, type the file name **My Friends** and click **Save**.
3	Close the workbook. Click **Open** and open **My Friends** again.
4	Click **Print Preview**. **Print** it only if your teacher says so.

DO IT - PRACTICAL 3 (WORKBOOK ACTIVITY 2.3)

1	Type **Grade 9 ICT Clas**. Click **Spelling** and correct the mistake.
2	Click **Copy**, click another cell and click **Paste**.
3	Click **Cut**, click a third cell and click **Paste**. The text moves!
4	Click **Undo**. What happens? Now click **Redo**. Save your work.

"Accountants once added with pencils. Today one formula does it!"

- THE KING OF ICT

A Fill in the blanks

8 marks

Use the words in the box.

WORD BANK

accounting	accountants	data	tabular
cells	charting	workbook	worksheet

1. A spreadsheet is an electronic version of the paper based ..... sheets.
2. Those paper sheets were used by ..... in the past.
3. A spreadsheet is used for the organization, analysis and storage of ..... .
4. An electronic spreadsheet keeps data in a ..... form.
5. Data are entered in the ..... of a table.
6. A spreadsheet gives the facility of functions, formulas, sorting and ..... .
7. A ..... may contain multiple worksheets.
8. Each sheet tab at the bottom of the window is one ..... .

B Name the tool

10 marks

Write the name of each tool. Use the words in the box.

WORD BANK

New	Open	Save	Print Preview	Print	Undo
Redo	Cut	Copy	Paste	Spelling	

1. .....
2. .....
3. .....
4. .....
5. .....
6. .....
7. .....
8. .....
9. .....
10. .....

C True or false

6 marks

Write T or F in the box.

1.		A spreadsheet is an electronic version of paper based accounting sheets.
2.		An electronic spreadsheet stores data in a tabular form.
3.		A workbook can have only one worksheet.
4.		Print Preview prints the document on paper.
5.		Redo does an undone action again.
6.		Copy removes an item from its current location.

D Match them

6 marks

Match the tool with what it does. Write the letter in the box.

A - Tool		B - What it does
1. New		**A.** To remove an item from its current location.
2. Open		**B.** To check the accuracy of spelling and grammar.
3. Print Preview		**C.** To create a new workbook.
4. Undo		**D.** To cancel a performed action.
5. Cut		**E.** To open a saved document.
6. Spelling		**F.** To see how a document will look if printed.

E Which tool would you use?

6 marks

Write the name of the tool.

You want to ...	Tool
see how your worksheet will look on paper before you print it	
keep your workbook in the computer to use it again later	
work again on a workbook that you saved yesterday	
cancel the last thing you did	
move a word from one cell to another cell	
check the spelling of the words in your worksheet	

F Choose the correct answer

6 marks

Underline your answer.

1. Which tool creates a new workbook?
   (a) Open (b) New (c) Save (d) Print
2. A workbook may contain
   (a) only one worksheet (b) multiple worksheets
   (c) no cells (d) only charts
3. Which tool cancels a performed action?
   (a) Redo (b) Cut (c) Undo (d) Paste
4. In a spreadsheet, data are entered in
   (a) cells (b) sheet tabs (c) charts (d) icons
5. Which tool pastes a copied or cut item?
   (a) Copy (b) Paste (c) Spelling (d) Open
6. Which of these is NOT a facility of a spreadsheet?
   (a) functions (b) sorting (c) charting (d) cooking

"Save early, save often. A power cut cannot take your saved work."

- THE KING OF ICT

CHAPTER 2

Inside a Worksheet

? WHERE IS G7?

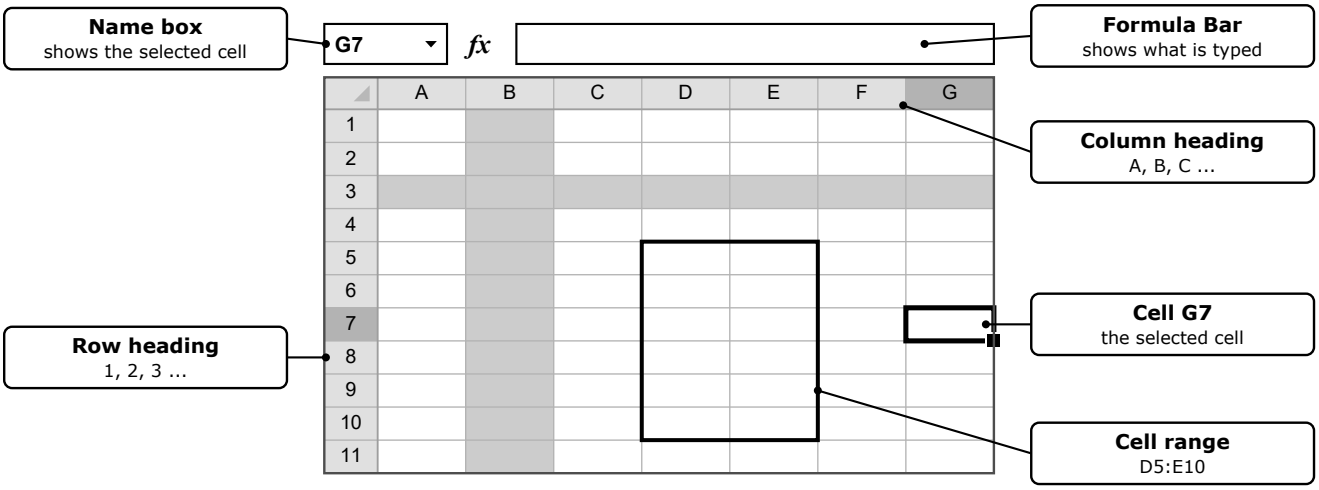
"Our teacher said: type your marks in cell G7. But where is G7?"

Go down column **G** and across row **7**. Where they meet is cell **G7**. This chapter shows the parts of a worksheet: columns, rows, cells, the Name box and the Formula Bar.



2.4 The parts of a worksheet

These are the special features of a spreadsheet (textbook page 24). In the picture, column B and row 3 are selected (grey).



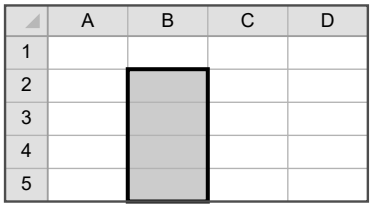
Part	What it is
Name box	Displays the address of the currently selected cell (here G7).
Formula Bar	Shows what is being typed in a cell.
Column	A line of cells from top to bottom. Its label begins with the letter A: A, B, C ...
Row	A line of cells from left to right. Its label starts with the number 1: 1, 2, 3 ...
Cell	The intersection of a row and a column, like G7.
Cell range	A range of selected cells in close proximity, like D5:E10.

2.5 Cell addresses

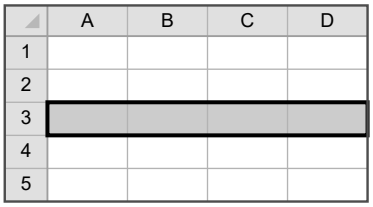
- Every cell has an **address**: the **column letter** first, then the **row number**. The cell in column G and row 7 is **G7**.
- The selected cell has a **thick border**. The **Name box** shows its address, and its column heading and row heading are shaded.
- Columns: A, B, C ... (the first one is A). Rows: 1, 2, 3 ... (the first one is 1).

2.6 Cell ranges and selecting data

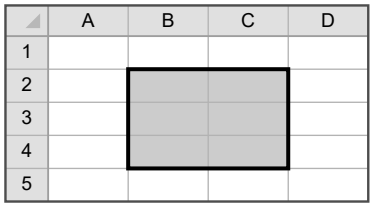
A **cell range** is a range of selected cells in close proximity (next to each other). Write the first cell, a colon (:) and the last cell. **D5:E10** is 2 columns and 6 rows: 12 cells.



B2:B5 - a range down a column: 4 cells



A3:D3 - a range along a row: 4 cells



B2:C4 - 2 columns and 3 rows: 6 cells

To select ...	Do this
One cell	Click the cell.
A cell range	Drag from the first cell to the last cell.
A whole column	Click its column heading (B).
A whole row	Click its row heading (3).

DO IT - PRACTICAL 4 (WORKBOOK ACTIVITY 2.4)

1	Click cell **G7**. The **Name box** shows G7.
2	Type **Kandy**. Watch the **Formula Bar** while you type. Press **Enter**.
3	Click the column heading **B**, then the row heading **3**.
4	Drag from **D5** to **E10** to select the cell range **D5:E10**.

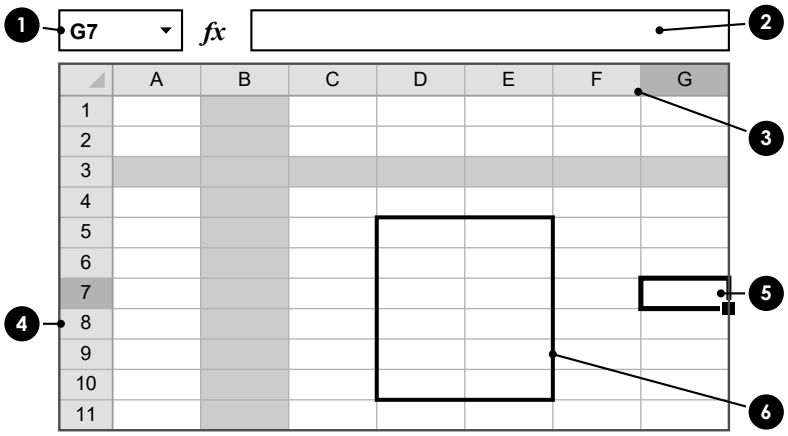
A Label the worksheet

6 marks

Write the name of each numbered part. Use the words in the box.

WORD BANK

Name box	Formula Bar	Sheet tab	Column heading
Row heading	Cell	Cell range	Title Bar



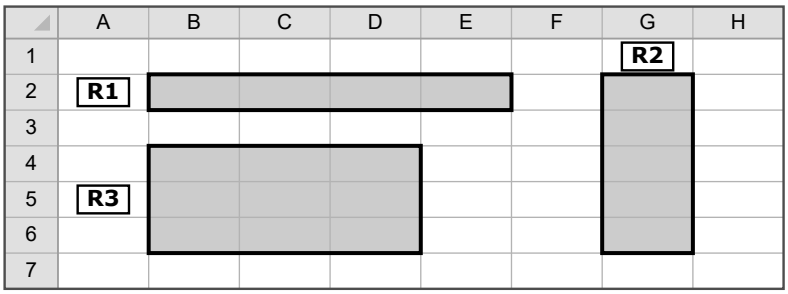
1. ..... 2. ..... 3. .....

4. ..... 5. ..... 6. .....

B Ranges in a worksheet

6 marks

Look at the shaded cell ranges R1, R2 and R3.



1. Write the cell range R1. .....
2. Write the cell range R2. .....
3. Write the cell range R3. .....
4. How many cells are there in R1? .....
5. How many cells are there in R3? .....
6. Which range (R1, R2 or R3) goes down a column? .....

C Fill in the blanks

8 marks

Use the words in the box.

WORD BANK

cell typed A 1 column range heading number

1. The Name box displays the currently selected ..... .
2. The Formula Bar shows what is being ..... in a cell.
3. The label of the first column is the letter ..... .
4. The label of the first row is the number ..... .
5. A cell is the intersection of a row and a ..... .
6. Selected cells in close proximity make a cell ..... .
7. To select a whole column, click its column ..... .
8. A cell address has the column letter first and then the row ..... .

D Read the worksheet

6 marks

The class list of Grade 9 B.

	A	B	C
1	**Name**	**House**	**Marks**
2	Nimal	Vijaya	72
3	Fathima	Gemunu	85
4	Kasun	Tissa	64
5	Tharushi	Parakrama	90
6	Ravi	Vijaya	58

1. What is in cell A4? .....
2. What is in cell C5? .....
3. Write the address of the cell with Gemunu. .....
4. Write the cell range of all the marks. .....
5. You click on the mark of Ravi. What does the Name box show? .....
6. You click on the mark of Tharushi and type 95. What does the Formula Bar show? .....

E Write the cell addresses

8 marks

Write the address of the cell of each shape.

	A	B	C	D	E	F	G
1					●		
2		★					
3				♦			
4							■
5	▲				+		
6			○			♥	

★ ..... ● .....

■ ..... ▲ .....

♦ ..... ♥ .....

○ ..... + ...

F True or false

6 marks

Write T or F in the box.

1. [ ] The cell address G7 means column G and row 7.

2. [ ] In a cell address, the row number comes first.

3. [ ] The Name box shows what is being typed in a cell.

4. [ ] To select a whole row, click its row heading.

5. [ ] The cell range B2:B5 has 4 cells.

6. [ ] The cell range A1:C2 has 5 cells.

G Choose the correct answer

4 marks

Underline your answer.

1. Which cell is in column D and row 4?
   (a) 4D (b) D4 (c) DD4 (d) D:4
2. How many cells are there in the range A1:B3?
   (a) 2 (b) 3 (c) 5 (d) 6
3. The Name box shows
   (a) what is typed (b) the selected cell (c) the sheet name (d) the chart
4. To select a whole row, click
   (a) the row heading (b) the Name box (c) the Formula Bar (d) a sheet tab

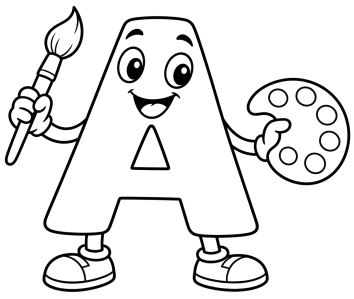
CHAPTER 3

Formatting the Cells

? A DULL RESULTS SHEET

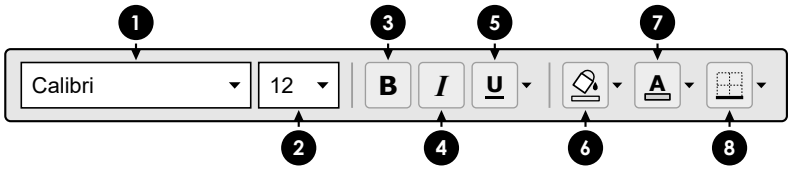
"My sports meet results sheet has all the right numbers, but it looks so dull. Nobody reads it!"

Format it! Fonts, sizes, colours, borders, alignment and number formats make a worksheet clear and attractive.



2.7 Cell formatting tools

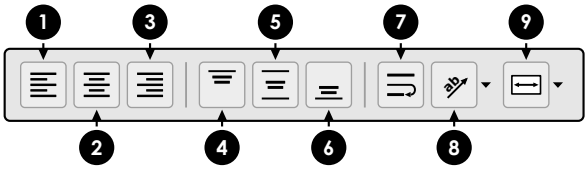
Select the cells first, then click a tool (textbook page 25):



No	Tool	What it does	Example
1	**Font**	Changes the type of the letters.	Times New Roman, Cambria, Nirmala UI
2	**Size**	Changes the size of the letters.	**Heading** **Sub-heading** normal
3	**Bold**	Makes the letters darker than normal.	**Sri Lanka**
4	**Italic**	Makes the letters slanted.	*Sri Lanka*
5	**Underline**	Draws a line beneath the letters.	Sri Lanka
6	**Fill colour**	Changes the cell background colour.	Sri Lanka
7	**Color**	Adds colours to the letters.	COLOUR ← colour it!
8	**Borders**	Applies borders (lines) to the cells.	

2.8 Alignment and positioning tools

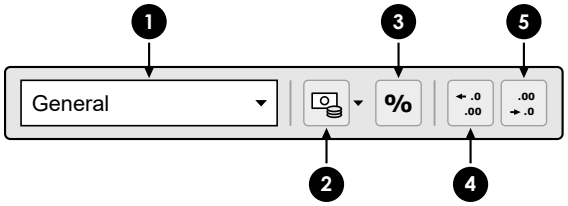
These tools place, wrap, turn and merge the content of a cell (page 25):



No	Tool	What it does	Example
1	**Left align**	Aligns the content to the left.	ICT
2	**Centre align**	Aligns the content to the middle (centre).	ICT
3	**Right align**	Aligns the content to the right.	ICT
4	**Top**	Aligns the content to the top of the cell.	ICT
5	**Middle**	Aligns the content to the middle, between top and bottom.	ICT
6	**Bottom**	Aligns the content to the bottom of the cell.	ICT
7	**Wrap Text**	Makes all the content visible within a cell.	All the words are seen
8	**Orientation**	Changes the direction of the text.	ICT
9	**Merge and Center**	Combines adjacent cells into one larger cell and aligns the content to the centre.	Sports Meet

2.9 Number tools

A number can be shown in many ways. The value in the cell stays the same; only the way it is shown changes (textbook page 26):



No	Tool	What it does
1	**Number Format box**	Shows **General**. Click it to choose a format from the list.
2	**Currency**	Shows a number as money: Rs. 10.00.
3	**Percentage**	Shows a number as a percentage: 0.25 is 25%.
4	**Increase decimals**	Shows more decimal places: 2.5 becomes 2.50.
5	**Decrease decimals**	Shows fewer decimal places: 2.50 becomes 2.5.

Click the **Number Format** box (General) and choose a format from the list:

	Format	It shows ...	Example (textbook page 26)
12	**Number**	numbers	**10, 25, 100, ...**
	Currency	money	**Rs. 10.00 $ 10.00 ¥ 10.00**
	Accounting	money; the currency symbols and decimal points are lined up in a column	**Rs. 10.00**
	Short Date	a date as MM/DD/YYYY (month / day / year)	**02/25/2018**
	Long Date	a detailed date	**Sunday, February 25, 2018**
	Time	a time	**1:30 p.m., 10:00 a.m.**
%	**Percentage**	a percentage	**23%, 45%, 75%**
½	**Fraction**	a fraction	**2/9, 4/5**
10^2	**Scientific**	scientific notation	**5677 = 5.68E+03**

DO IT - PRACTICAL 5 (WORKBOOK ACTIVITY 2.5)

1	Type **Sports Meet Results** in A1 and four house names in A2:A5.
2	Click A1: **Font** Cambria, **Size** 16, **Bold** and a **Fill colour**.
3	Select A2:A5: **Italic**, **Underline** and a red font **Color**.
4	Select A1:B5. Click **Borders** and choose **All Borders**.

DO IT - PRACTICAL 6 (WORKBOOK ACTIVITY 2.6)

1	Type **Sports Meet** in A1. Select A1:D1. Click **Merge and Center**.
2	Type **ICT** in B3. Try **Left**, **Centre** and **Right align**.
3	Make row 3 taller. Try **Top**, **Middle**, **Bottom** and **Orientation**.
4	Type a long sentence in B5 and click **Wrap Text**.

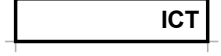
DO IT - PRACTICAL 7 (WORKBOOK ACTIVITY 2.7)

1	Type **10**, **0.25**, **2/25/2018**, **13:30** and **5677** in A1 to A5.
2	A1: click **Currency**, then **Increase** and **Decrease decimals**.
3	A2: click **Percentage**. A3: choose **Long Date**, then **Short Date**.
4	A4: choose **Time**. A5: choose **Scientific**. It shows **5.68E+03**.

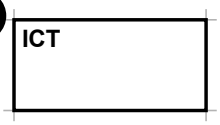
A Which tool was used?

8 marks

Each cell shows the work of one alignment or positioning tool. Name it.

1. 

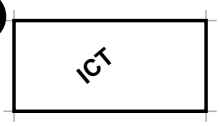
2. 

3. 

4. 

5. 

6. 

7. 

8. 

B What will the cell show?

6 marks

Write what the cell shows after the format.

The value	Number format	The cell shows
0.75	Percentage	
12500	Scientific	
250	Currency	
0.5	Fraction	
3/10/2025	Long Date	
7.25	Decrease decimals (click once)	

C True or false

6 marks

Write T or F in the box.

1. ☐ Italic makes the letters slanted.
2. ☐ Fill colour changes the colour of the letters.
3. ☐ Wrap Text makes all the content visible within a cell.
4. ☐ Merge and Center combines adjacent cells into one larger cell.
5. ☐ Orientation changes the size of the letters.
6. ☐ The Short Date format shows a date as DD/MM/YYYY.

D Name the tools

8 marks

Write the name of each tool. Use the words in the box.

WORD BANK

Bold	Italic	Underline	Fill colour	Color
Borders	Wrap Text	Merge and Center	Orientation	Size

1 

2 

3 

4 

5 

6 

7 

8 

E Which number format?

7 marks

Write the number format of each cell.

The cell shows	Number format
Rs. 75.00	
35%	
3/4	
4.50E+06	
08/15/2025	
Friday, August 15, 2025	
10:00 a.m.	

F Which tool would you use?

8 marks

Write the name of the tool.

You want to ...	Tool
make the heading darker than normal	
change the background colour of a cell	
draw lines around the cells of a table	
make the letters slanted	
join A1:E1 into one cell with the heading in the centre	
show a long sentence completely inside one cell	
change the colour of the letters	
change the direction of the text	

G Choose the correct answer

4 marks

Underline your answer.

1. Which tool makes letters darker than normal?
   (a) Italic (b) Bold (c) Underline (d) Borders
2. Which tool changes the background colour of a cell?
   (a) Color (b) Fill colour (c) Size (d) Font
3. 0.45 with the Percentage format shows
   (a) 0.45% (b) 45% (c) 4.5% (d) 450%
4. Which format shows 2/25/2018 as Sunday, February 25, 2018?
   (a) Short Date (b) Time (c) Long Date (d) Number

CHAPTER 4

Formulas and Functions

? THE CLASS TRIP FUND

"We save money for our class trip every month. How much do we have now? Which month was the best?"

Type the amounts once. **Formulas** and **functions** find the total, the average, the highest and the lowest in a moment - and they never make a mistake in adding.



2.10 Formulas

A **formula** does a calculation. Every formula begins with the **=** sign. Then come numbers or cell addresses and these symbols (textbook page 27):

Symbols used in a spreadsheet formula

Symbol	Name	Meaning
=	equals	Used before a formula or function
+	plus	Add
-	minus	Subtract
*	star	Multiply
/	slash	Divide

Formulas with numbers

Formula	Result
=25+15	**40**
=25-15	**10**
=25*4	**100**
=100/4	**25**

Use cell addresses. A1 has 25 and B1 has 5. C1 has **=A1+B1**: the **cell** shows the result, and the **Formula Bar** shows the formula. Change A1, and the result changes by itself!

C1 *fx* =A1+B1

	A	B	C	D
1	25	5	30	
2				

Formula in C1	C1 shows	Now A1 = 40
=A1+B1	30	45
=A1-B1	20	35
=A1*B1	125	200
=A1/B1	5	8

	A	B	C	D
1	**Item**	**Price (Rs.)**	**Quantity**	**Amount (Rs.)**
2	Exercise book	120	5	=B2*C2
3	Pen	25	4	=B3*C3
4	Pencil	15	6	=B4*C4
5	Ruler	40	2	=B5*C5
6	**Total**			=D2+D3+D4+D5

The stationery shop bill: what you type

The stationery shop bill. The amount is the price multiplied by the quantity, so D2 has **=B2*C2**.

D2 to D5 show **600, 100, 90, 80**. The total in D6 is **870**.

DO IT - PRACTICAL 8 (WORKBOOK ACTIVITY 2.8)

1	Type **25** in A1, **5** in B1 and **=A1+B1** in C1. C1 shows **30**.
2	Type **=A1-B1**, **=A1*B1** and **=A1/B1** in C2, C3 and C4.
3	Click C1 and read the Formula Bar. Change A1 to **40**. Watch C1:C4.
4	Make the stationery shop bill of 2.10 with its formulas. Save it.

2.11 Functions

A **function** is a ready-made formula. Type =, the name of the function and the cell range in brackets: **=SUM(B2:B6)** adds B2, B3, B4, B5 and B6. SUM, AVERAGE, MAX and MIN are some basic functions:

Function	Example	What it does
SUM	**=SUM(B2:B6)**	To find the total of a range of cells.
AVERAGE	**=AVERAGE(B2:B6)**	To find the average of a cell range.
MAX	**=MAX(B2:B6)**	To find the highest value in a cell range.
MIN	**=MIN(B2:B6)**	To find the lowest value in a cell range.

	A	B
1	**Month**	**Amount (Rs.)**
2	January	2500
3	February	3200
4	March	2800
5	April	4100
6	May	3400
7	**Total**	=SUM(B2:B6)
8	**Average**	=AVERAGE(B2:B6)
9	**Highest**	=MAX(B2:B6)
10	**Lowest**	=MIN(B2:B6)

The class trip fund: what you type

	A	B
1	**Month**	**Amount (Rs.)**
2	January	2500
3	February	3200
4	March	2800
5	April	4100
6	May	3400
7	**Total**	16000
8	**Average**	3200
9	**Highest**	4100
10	**Lowest**	2500

... and what you see

Type the amounts once. The functions find the total, the average, the highest and the lowest. Change an amount, and all four results change.

DO IT - PRACTICAL 9 (WORKBOOK ACTIVITY 2.9)

1	Make the class trip fund worksheet of 2.11 (A1:B6).
2	Type **=SUM(B2:B6)** in B7. It shows **16000**.
3	Type the AVERAGE, MAX and MIN functions in B8, B9 and B10.
4	Change the March amount to **3300**. Watch B7:B10 change.

A Match the symbols

5 marks

Write the correct letter in the box.

A - Symbol		B - Meaning
1. =		**A.** Divide
2. +		**B.** Used before a formula or function
3. -		**C.** Multiply
4. *		**D.** Add
5. /		**E.** Subtract

B Write the formulas

8 marks

The school canteen bill. 1 mark for the formula, 1 mark for the result.

	A	B	C	D
1	**Item**	**Price (Rs.)**	**Quantity**	**Amount (Rs.)**
2	Bun	60	3	
3	Cup of tea	40	2	
4	Fish patty	70	4	
5	Banana	25	6	
6	**Total**			
7				

1. D2: the amount for the buns

= ..... Result: .....

2. D4: the amount for the fish patties

= ..... Result: .....

3. D6: the total of D2 to D5 (use + signs)

= ..... Result: .....

4. B7: how much more a fish patty costs than a bun

= ..... Result: .....

C Find the results

8 marks

The test marks of five students. Use the functions.

	A	B	C	D
1	**Name**	**ICT**	**Maths**	**Science**
2	Nimal	65	72	58
3	Fathima	80	64	77
4	Kasun	45	59	62
5	Tharushi	92	85	88
6	Ravi	70	48	66

Function	Result
=SUM(B2:B6)	
=AVERAGE(B2:B6)	
=MAX(C2:C6)	
=MIN(D2:D6)	
=SUM(B2:D2)	
=AVERAGE(B2:D2)	
=MAX(B2:D6)	
=MIN(B2:D6)	

D Results from a worksheet

8 marks

Use the values in the worksheet.

	A	B	C
1	12	4	3
2	20	5	8
3	6	10	2

Formula	Result
=A1+B1	
=A2-C2	
=B2*C1	
=A2/B2	

Formula	Result
=A1+B1+C1	
=B3*C3	
=(A3+B3)/C3	
=(A2-B3)*C1	

E Which function?

6 marks

Tick the function you would use.

You want to find ...	SUM	AVERAGE	MAX	MIN
The total money collected for the school trip	☐	☐	☐	☐
The highest mark in the ICT test	☐	☐	☐	☐
The average rainfall of a month	☐	☐	☐	☐
The lowest price in a price list	☐	☐	☐	☐
The total runs of a cricket team	☐	☐	☐	☐
The average age of the students of a class	☐	☐	☐	☐

F Work them out

8 marks

Write the result of each formula.

1. **=15+27** .....
2. **=50-18** .....
3. **=6*8** .....
4. **=72/8** .....
5. **=(4+6)*5** .....
6. **=(20-8)/4** .....
7. **=2*(15+5)** .....
8. **=(9+11)/(7-2)** .....

G Spot the mistakes

4 marks

Kasun made FOUR mistakes. Underline them. Write the correct words below.

Kasun wrote: "Every formula begins with a plus (+) sign. The symbol for multiply is x. =AVERAGE(B2:B6) finds the total of B2 to B6. MAX finds the lowest value in a cell range."

1. ..... 2. ..... 3. ..... 4. .....

CHAPTER 5

Charts

? WHO IS WINNING?

"Forty numbers on the sports meet board, and I still cannot see which house is winning!"

Show the numbers in a **chart**. A chart shows spreadsheet data graphically: one look is enough to compare them.



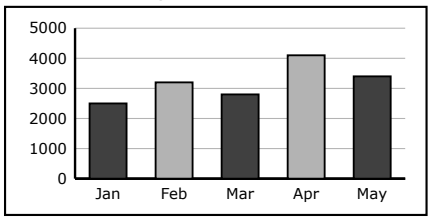
2.12 Charts

Charts can be used to display spreadsheet data graphically, so that we can analyse the data. The chart buttons are in the **Charts** group of the **Insert** tab (textbook page 27):

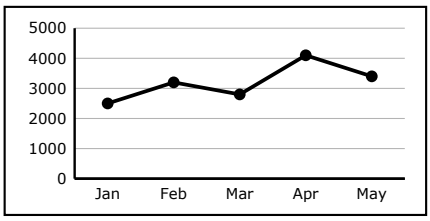


To create a chart: select the data (for example A1:B6 of the trip fund), click the **Insert** tab, click a chart in the **Charts** group (Column, Line, Pie ...) and choose one picture of that chart. In LibreOffice Calc: select the data, then **Insert** and **Chart**.

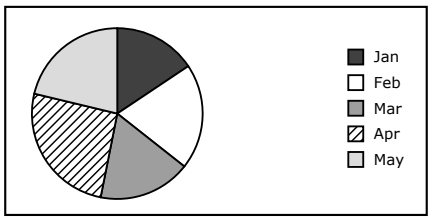
The class trip fund of 2.11 in six kinds of chart:



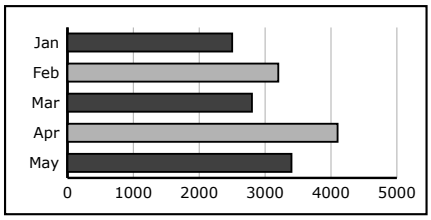
Column chart: columns (bars) that stand up



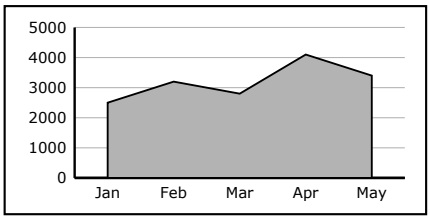
Line chart: points joined by a line



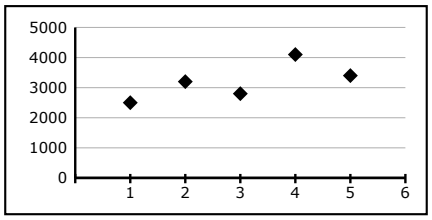
Pie chart: a circle cut into slices



Bar chart: bars that lie from left to right



Area chart: a line with the area under it filled



Scatter chart: points (dots) only, not joined

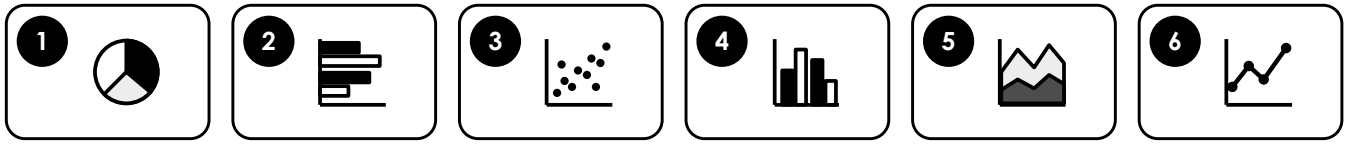
DO IT - PRACTICAL 10 (WORKBOOK ACTIVITY 2.10)

1	Select the trip fund data **A1:B6**. Click the **Insert** tab.
2	In **Charts** click **Column** and choose the first chart.
3	Make a **Line**, **Pie**, **Bar**, **Area** and **Scatter** chart too.
4	Which chart shows the data best? Why? Save your work.

A Name the charts

6 marks

Write the name of each chart.



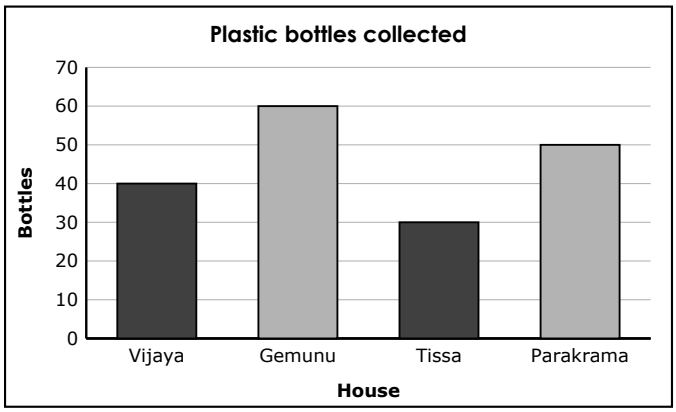
"A good chart tells the whole story in one look."

- THE KING OF ICT

B Read the chart

6 marks

The houses collected plastic bottles for recycling.



1. What type of chart is this? .....
2. Which house collected the most bottles? .....
3. Which house collected the fewest bottles? .....
4. How many bottles did Vijaya collect? .....
5. How many more bottles did Gemunu collect than Tissa? .....
6. Which function finds the total number of bottles? What is the total? .....

C Choose the correct answer

4 marks

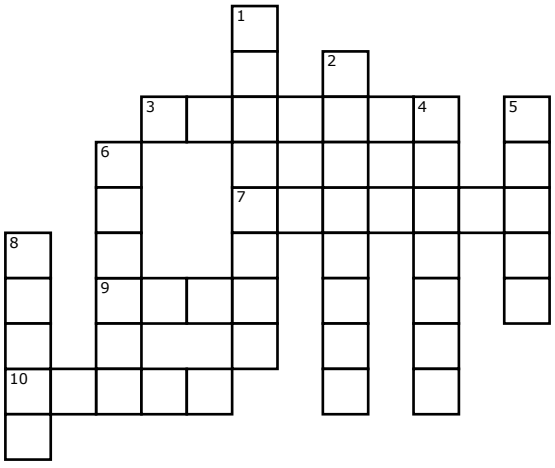
Underline your answer.

1. Charts display spreadsheet data
   (a) as sounds (b) graphically
   (c) as formulas (d) as cell addresses
2. Which chart is a circle cut into slices?
   (a) Column chart (b) Line chart (c) Pie chart (d) Scatter chart
3. In a bar chart, the bars
   (a) stand up (b) lie from left to right
   (c) are slices (d) are dots
4. Which of these is NOT a chart button?
   (a) Column (b) Area (c) Scatter (d) Borders

D Crossword

10 marks

All the words are in this unit. Write one letter in each square.



ACROSS

3. It always begins with = (7)
7. This tool draws lines around cells (7)
9. It cancels a performed action (4)
10. D5:E10 is a cell ... (5)

DOWN

1. A spreadsheet file; it may contain multiple worksheets (8)
2. The number format for money (8)
4. The function that finds the average of a cell range (7)
5. It puts a copied or cut item in a new place (5)
6. Its label begins with the letter A (6)
8. It shows spreadsheet data graphically (5)

"Do your best work today. Tomorrow you will be proud of it."

- THE KING OF ICT

REVISION

Unit 2 in One Page

Read this the night before the test

Topic	The key facts
Spreadsheet	An electronic version of the paper based accounting sheets. It organizes, analyses and stores data in a tabular form: functions, formulas, sorting and charting.
Workbook	A spreadsheet file. A workbook may contain multiple worksheets (sheet tabs).
Common tools	New, Open, Save, Print Preview, Print, Undo, Redo, Cut, Copy, Paste, Spelling.
The window	Name box = the selected cell. Formula Bar = what is typed. Columns A, B, C ... Rows 1, 2, 3 ...
Cells	A cell is where a row and a column meet. Address = column letter + row number: G7.
Cell range	Selected cells close together: first cell, a colon, last cell. D5:E10 has 12 cells.
Selecting	Click a cell; drag over a range; click a column heading or a row heading.
Formatting	Font, Size, Bold, Italic, Underline, Fill colour (background), Color (letters), Borders.
Alignment	Left, Centre and Right align; Top, Middle and Bottom; Wrap Text; Orientation; Merge and Center.
Number formats	Number, Currency, Accounting, Short Date (MM/DD/YYYY), Long Date, Time, Percentage, Fraction, Scientific (5.68E+03). Increase and decrease decimals.
Formulas	Begin with =. Add +, subtract -, multiply *, divide /. =A1+B1 is updated when A1 changes.
Functions	SUM (total), AVERAGE, MAX (highest), MIN (lowest) of a cell range: =SUM(B2:B6).
Charts	Show data graphically: column, line, pie, bar, area, scatter. Select the data, Insert, choose.

TICK WHEN YOU CAN DO IT

- [ ] I can say what a spreadsheet, a workbook and a worksheet are.
- [ ] I can use New, Open, Save, Print Preview, Print, Undo, Redo, Cut, Copy, Paste and Spelling.
- [ ] I can find the Name box, the Formula Bar, columns, rows, cells and cell ranges.
- [ ] I can write cell addresses (G7) and cell ranges (D5:E10).
- [ ] I can format cells with the formatting, alignment and number tools.
- [ ] I can write formulas with = + - * / and use SUM, AVERAGE, MAX and MIN.
- [ ] I can create column, line, pie, bar, area and scatter charts.

WORD POWER

Unit 2 Word List

Twenty-four words you must know

Word	What it means
Electronic spreadsheet	A computer application to organize, analyse and store data in a table of cells.
Workbook	A spreadsheet file. It may contain multiple worksheets.
Worksheet	One sheet of cells in a workbook. Each worksheet has a sheet tab.
Column	A line of cells from top to bottom, labelled A, B, C ...
Row	A line of cells from left to right, labelled 1, 2, 3 ...
Cell	The intersection of a row and a column.
Cell address	The column letter and the row number of a cell, like G7.
Cell range	Selected cells in close proximity, like D5:E10.
Name box	Displays the address of the currently selected cell.
Formula Bar	Shows what is being typed in a cell.
Print Preview	Shows how a document will look if printed.
Undo / Redo	Cancels a performed action / does an undone action again.
Cut / Copy / Paste	Removes an item / copies it / puts the cut or copied item in a new place.
Font / Size	The type of the letters / the size of the letters.
Fill colour	Changes the background colour of a cell.
Borders	Lines drawn around cells.
Wrap Text	Makes all the content visible within a cell.
Merge and Center	Combines adjacent cells into one larger cell and centres the content.
Orientation	Changes the direction of the text.
Number format	The way a number is shown: Currency, Percentage, Date, Time, Fraction ...
Formula	A calculation that begins with =, like =A1+B1.
Function	A ready-made formula: SUM, AVERAGE, MAX, MIN.
Sorting	Putting data in order, for example from the smallest to the largest.
Chart	A picture of data: column, line, pie, bar, area or scatter.

"Learn the words today, and the formulas will follow tomorrow."

- THE KING OF ICT

END OF UNIT

Unit 2 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. A spreadsheet is an electronic version of
   (a) paper based accounting sheets (b) a story book
   (c) a drawing book (d) a dictionary
2. Which tool cancels a performed action?
   (a) Redo (b) Undo (c) Cut (d) Paste
3. The Name box displays
   (a) what is typed in a cell (b) the currently selected cell
   (c) the sheet name (d) the chart type
4. How many cells are there in the range B2:C5?
   (a) 4 (b) 6 (c) 8 (d) 10
5. Which tool makes all the content visible within a cell?
   (a) Wrap Text (b) Orientation (c) Borders (d) Fill colour
6. In which format is 5677 shown as 5.68E+03?
   (a) Currency (b) Fraction (c) Scientific (d) Percentage
7. A1 has 5, A2 has 10 and A3 has 15. What does =SUM(A1:A3) give?
   (a) 15 (b) 30 (c) 25 (d) 5
8. A - Every formula begins with =. B - * is the symbol for multiply. C - MIN finds the highest value. Which are correct?
   (a) A, B only (b) B, C only (c) A, C only (d) A, B and C

2 Fill in the blanks

8 marks

One word, number or sign in each blank.

1. A workbook may contain multiple ..... .
2. The label of the first column is the letter ..... .
3. The intersection of a row and a column is a ..... .
4. The ..... shows what is being typed in a cell.
5. The ..... tool changes the background colour of a cell.
6. Every formula begins with the ..... sign.
7. If A1 = 4, A2 = 9 and A3 = 2, then =MAX(A1:A3) gives ..... .
8. The symbol for divide is ..... .

3 Name the tools

8 marks

Write the name of each tool or chart button.

1 

2 

3 

4 

5 

6 

7 

8 

4 Answer these

16 marks

The sports meet points of the four houses. 2 marks each.

	A	B	C	D
1	**House**	**Track**	**Field**	**Total**
2	Vijaya	85	64	
3	Gemunu	92	58	
4	Tissa	70	75	
5	Parakrama	66	81	
6				
7				

1. Write a formula for D2 to add the track and field points of Vijaya. What does D2 show?

2. Write a function for B6 to find the total track points of all the houses. What does it show?

3. Write a function for B7 to find the average track points. What does it show?

4. Write a function to find the highest field points. What does it show?

5. Write the cell range of the field points. How many cells are in it?

6. You click on cell A3. What does the Name box show? What is in A3?

7. Name two tools you would use to make row 1 stand out as a heading row.

8. Which chart would you use to compare the total points of the houses? Write its first step.